1. $N(t) = 2000(1+3t)^3$

$N'(t) = 2000 \cdot 3(1+3t)^2 \cdot 3$

$= (2000 \cdot 3 \cdot 3)(1+3t)^2$

$= (6000 \cdot 3)(1+3t)^2$

$= \underline{18000}(1+3t)^2$

$N'(2) = 18000(1+3(2))^2$

$= 18000(7)^2$

$= 18000(49) = \underline{\underline{882000}}$

2. $U = 2n^2 + 1$

$U' = 2 \cdot 2n + 0 = \underline{4n}$

$V = 3n - 4$

$V' = 3 - 0 = \underline{3}$

$C'(n) = U'V + UV'$

$4n(3n-4) + (2n^2+1)(3)$

$(12n^2 - 16n) + (6n^2 + 3)$

$12n^2 - 16n + 6n^2 + 3$

$(12n^2 + 6n^2) - 16n + 3$

$\underline{18^2 - 16n + 3}$

soal ↑

$C(n) = (2n^2+1)(3n-4)$

$C'(3) = 18(3)^2 - 16(3) + 3$

$= 18(9) - 48 + 3$

$= 162 - 48 + 3$

$= 114 + 3 = \underline{117}$

JOYKO 30 Lines 6 mm

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3. $S(t) = (t^2+1)(2t-3)$

$u = t^2+1 \rightarrow u' = 2t$

$v = 2t-3 \rightarrow v' = 2$

$S'(t) = u'v + uv'$

$= 2t(2t-3) + (t^2+1)(2)$

$= (4t^2 - 6t) + (2t^2 + 2)$

$= 4t^2 - 6t + 2t^2 + 2$

$= (4t^2 + 2t^2) - 6t + 2$

$= 6t^2 - 6t + 2$

$S'(2) = 6(2)^2 - 6(2) + 2$

$= 6(4) - 12 + 2$

$= 24 - 12 + 2$

$= 12 + 2$

$= 14$

4. $k(t) = \dfrac{100t}{(t^2+4)}$

$u = 100t \rightarrow u' = 100$

$v' = t^2+4 \rightarrow v' = 2t$

$k'(t) = \dfrac{(u'v - uv')}{v^2}$

$= \dfrac{(100(t^2+4) - (100t)(2t))}{(t^2+4)^2}$

$= \dfrac{((100t^2 + 400) - 200t^2)}{(t^2+4)^2}$

$= \dfrac{(-100t^2 + 400)}{(t^2+4)^2}$

$k'(2) = \dfrac{(-100(2)^2 + 400)}{(2^2+4)^2}$

$= \dfrac{(-100(4) + 400)}{(4+4)^2}$

$= \dfrac{(-400 + 400)}{(8)^2} = \dfrac{0}{64} = 0$

5. $V = (\frac{4}{3})\pi r^3$ dan $r = 2t+1$

$V(t) = (\frac{4}{3})\pi(2t+1)^3$

$V'(t) = (\frac{4}{3})\pi \cdot 3(2t+1)^2 \cdot 2$

$= (\frac{4}{3} \cdot 3 \cdot 2)\pi(2t+1)$

$V'(t) = (4 \cdot 2)\pi(2t+1)^2$

$= 8\pi(2t+1)^2$

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$V'(1) = 8M(2(1)+1)^2$

$= 8M(2+1)^2$

$= 8M(3)^2$

$= 8M(9) = 72M$

$= 72 \times 3,1416 = \underline{226,19}$

6. $P(t) = 5t.\sqrt{(t+4)}$, cari $P'(5)$

$= \sqrt{(t+4)} = (t+4)^{(\frac{1}{2})}$

$u = 5t \rightarrow u' = 5$

$V = (t+4)^{(\frac{1}{2})} \rightarrow V' = (\frac{1}{2})(t+4)^{(\frac{-1}{2})} . 1$

$= \frac{1}{(2\sqrt{(t+4)})}$

$P'(t) = u'V + uV'$

$= \frac{5.\sqrt{(t+4)} + 5t \cdot 1}{(2\sqrt{(t+4)})} = \frac{5\sqrt{(t+4)} + 5t}{(2\sqrt{(t+4)})}$

$P'(5) = \frac{5\sqrt{(5+4)} + 5(5)}{(2\sqrt{(5+4)})} = \frac{5\sqrt{9} + 25}{(2\sqrt{9})} = \frac{5(3)+25}{(2.3)} = \frac{15+25}{6}$

$= \frac{90}{6} + \frac{25}{6} = \frac{115}{6} = \underline{19,17}$

7. $T(x) = \frac{500x}{(x+5)}$ cari $T'(5)$

$u = 500x \rightarrow u' = 500$

$V = x+5 \rightarrow V' = 1$

$T'(x) = \frac{(u'V - uV')}{V^2} = \frac{(500(x+5) - (500x)(1))}{(x+5^2)} = \frac{(500 + 2500 - 500x)}{(x+5)^2}$

$= \frac{2500}{(x+5)^2}$. $T'(5) = \frac{2500}{(5+5)^2} = \frac{2500}{(10)^2} = \frac{2500}{100} . 25$

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8. $T(t) = 30 + 4t^2 - \frac{t^3}{3}$ $T'(3)$

$30 \to 0$

$4t^2 \to 4 \cdot 2t^1 = 8t$

$-(\frac{1}{3})t^3 \to -(\frac{1}{3}) \cdot 3t^2 = -t^2$

$T'(t) = 0 + 8t - t^2$

$= 8t - t^2$

$T'(3) = 8(3) - (3)^2 = 24 - 9$

$= 15$

9. $B(x) = \frac{200x^2}{(x+1)}$ $B'(4)$

$U = 200x^2 \to U' = 200 \cdot 2x = 400x$

$V = x+1 \to V' = 1$

$B'(x) = \frac{(U'V - UV')}{V^2} = \frac{(400x(x+1) - (200x^2)(1))}{(x+1)^2} = \frac{(400x^2 + 400x) - 200x^2}{(x+1)^2}$

$= \frac{(400x^2 + 400x - 200x^2)}{(x+1)^2} = \frac{(200x^2 + 400x)}{(x+1)^2}$

$B'(4) = \frac{(200(4)^2 + (400(4))}{(4+1)^2} = \frac{(200(16)) + 1600)}{(5)^2} = \frac{(3200 + 1600)}{25} = \frac{4800}{25}$

$= 192$

10. $H(t) = (3t+1)^2(2-t)$ $h'(1)$

$U = (3t+1)^2 \to U' = 2(3t+1) \cdot 3 = 6(3t+1)$ Suku kedua

$V = 2-t \to V' = 0 - 1 = -1$

Suku pertama

$(3t+1)(2-t) = 6t - 3t^2 + 2 - t$

$= -3t^2 + 5t + 2$

$6(-3t^2 + 5t + 2) = -18t^2 + 30t + 12$

$(3t+1)^2 = 9t^2 + 6t + 1$

$(3t+1)^2(-1) = -9t^2 - 6t - 1$

$h'(t) = (-18t^2 + 30t + 12) + (-9t^2 - 6t - 1)$

$= (-18t^2 - 9t^2) + (30t - 6t) + (12 - 1)$

$= -27t^2 + 24t + 11$

Gabungan

$h'(1) = -27(1)^2 + 24(1) + 11$

$= -27(1) + 24 + 11$

$= -27 + 24 + 11$

$= -3 + 11 = 8$

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